Slew budget

Slew (transition) is what library cells require; long RC violates the budget

The idea

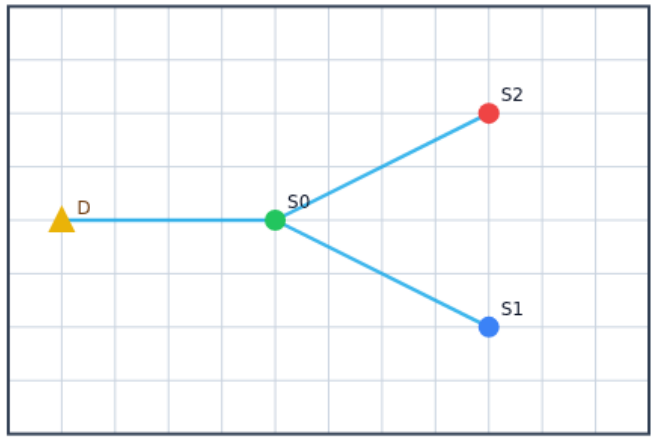
- Estimate_slew equals k times R times C_total with k about 2.2
- Slew_ok is true when slew is at most the limit, eight on the toy

Slew definition

SLEW BUDGET

Slew definition

Step 1 / 5 · def · module01-03-slew-budget



Explanation

Toy transition ≈ 2.2 · R · C_total on the segment.

- k≈2.2
- Not delay

```
estimateSlew
```

Slew definition

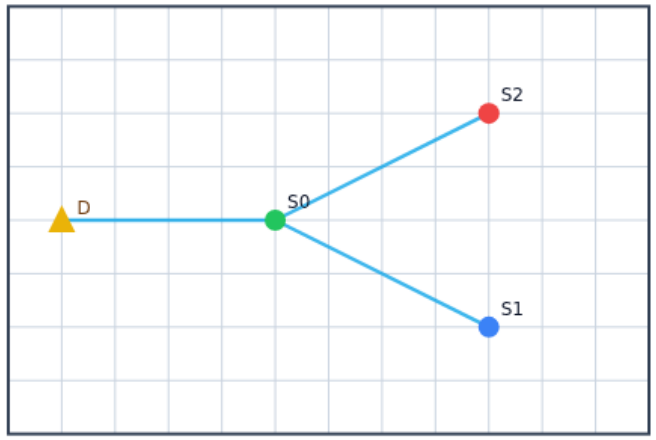
Limit eight

SLEW BUDGET

Limit eight

Step 2 / 5 · limit · module01-03-slew-budget

slew limit 8



Explanation

GOLDENS.slewLimit = 8 on tiny_buf.

- Pass/fail gate

```
limit=8
```

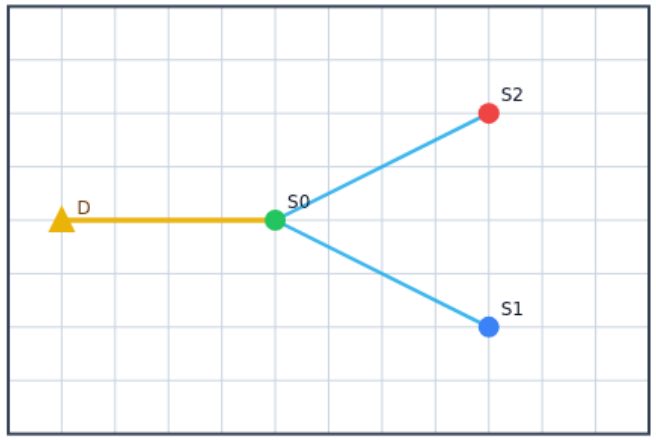
Limit eight

Hot edge

SLEW BUDGET

Hot edge

Step 3 / 5 · hot · module01-03-slew-budget



Explanation

Long e0 with branch load can approach or exceed the limit.

- Branch C

```
e0
```

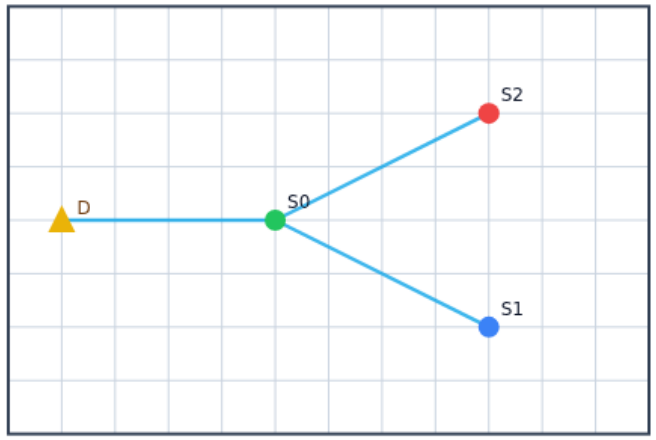
Hot edge

Pass/fail

SLEW BUDGET

Pass/fail

Step 4 / 5 · ok · module01-03-slew-budget



Explanation

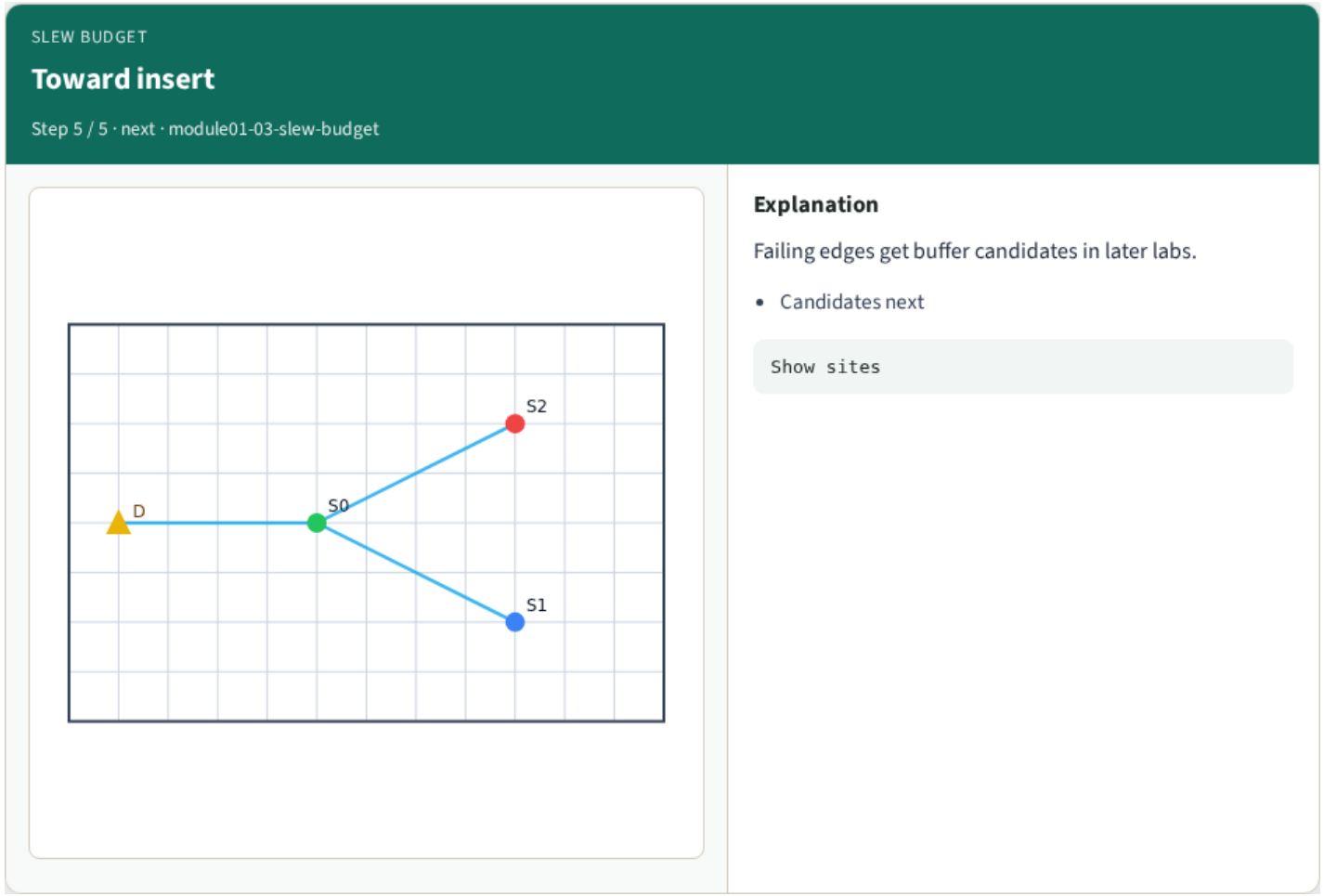
slew_ok gates whether slew-driven insertion fires.

- Boolean check

```
slewOk
```

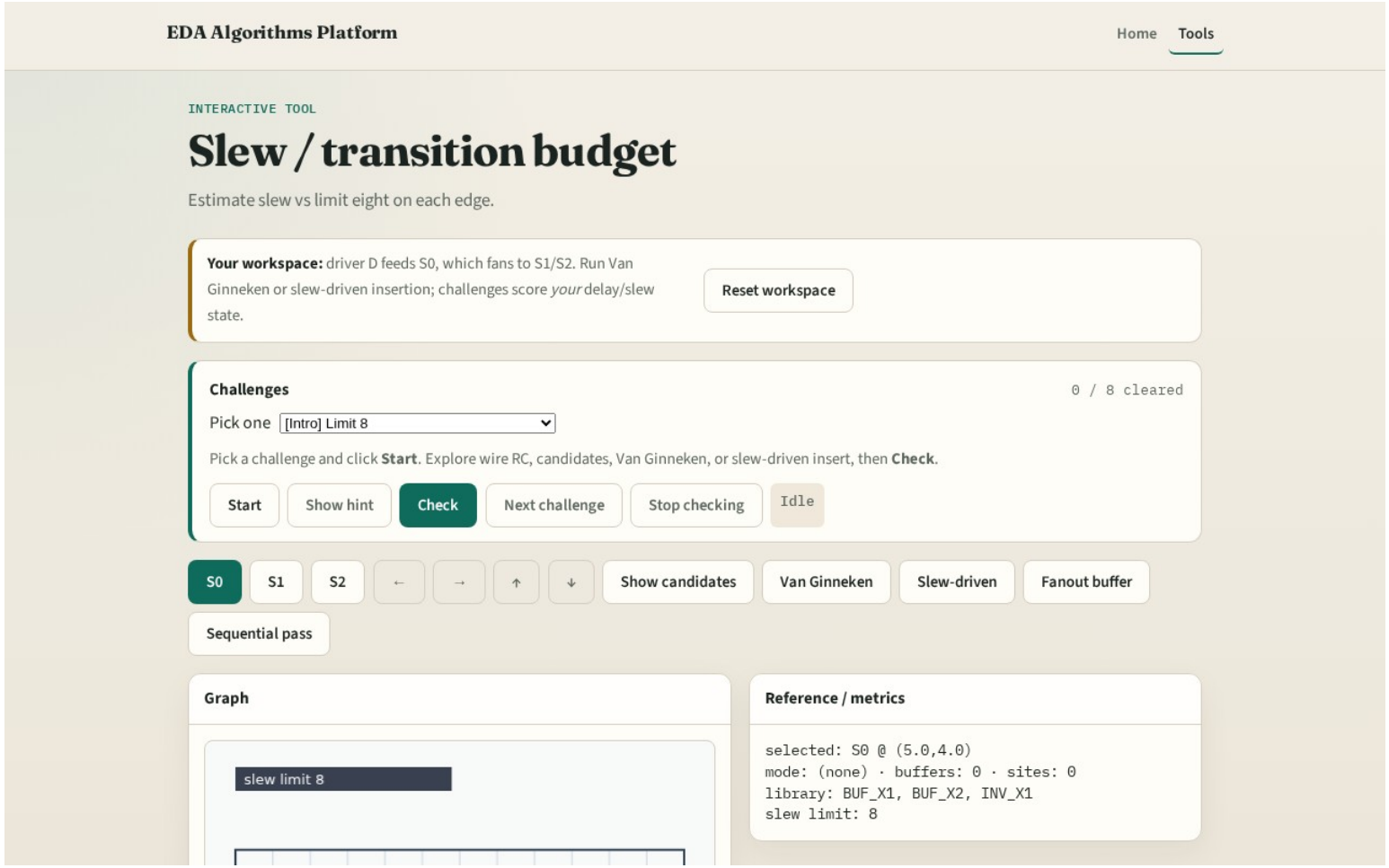
Pass/fail

Toward insert



Toward insert

Browser lab track



Browser lab starter

Implement track

- Call `estimate_slew` and `slew_ok` in solvers
- Assert limit from tiny_buf.json

Pitfalls

- Using delay as slew
- Ignoring C_load
- Hard-coding pass without measuring

Your turn

- Next: place candidate buffer sites along edges